

PROJECT SETU

AI-Powered Government Service Delivery, Intelligent Grievance Redressal & Multi-Departmental Governance Platform

Architecture: Full-Stack React + Node.js + PostgreSQL

Languages: 12 Indian Languages (100% Native i18n)

Security: JWT + RBAC + Audit Trails

1. Executive Summary & Mission

PROJECT SETU (*Smart E-Governance & Transparency Unit*) is an enterprise digital public infrastructure (DPI) platform engineered to bridge the gap between citizens, municipal field officers, department directors, and state administrators. Built strictly according to Government of India digital standards, it delivers automated grievance triaging, public service fulfillment, welfare scheme discovery, and transparent SLA oversight.

2. System Architecture & Tech Stack

Frontend Architecture

REACT 18 + VITE

Modern single-page application with TypeScript, Tailwind CSS, Lucide icons, and reactive i18next language engine.

Backend REST Engine

NODE.JS + EXPRESS

High-performance TypeScript REST API server with JWT authentication, role-based access control, and secure file handling.

Database & ORM

POSTGRESQL 14+ / PRISMA

Relational database storing citizen accounts, master taxonomy, grievances, applications, SLA timelines, and tamper-evident audit ledgers.

AI & NLP Intelligence

FASTAPI MICROSERVICE

Natural Language Processing model performing automated department routing, priority calculation, and anomaly surge detection.

3. The 4 Dedicated User Workbenches & Login Matrix

USER ROLE	OFFICIAL DEMO ACCOUNT	ROUTE	CORE CAPABILITIES & WORKFLOWS
 Public Citizen	<code>citizen@setu.gov.in</code>	<code>/citizen</code>	Lodge complaints, GPS photo evidence, 7-step direct service applications, welfare scheme eligibility checker, and SETU AI Assistant.
 Field Officer	<code>officer.water@setu.gov.in</code>	<code>/officer</code>	Assigned municipal workbench, on-site inspection queue, GPS map navigation, photo proof uploads, and case resolution.
 Govt Officer (HOD)	<code>senior.officer@setu.gov.in</code>	<code>/senior-officer</code>	Senior Executive command center, SLA overdue escalations (Level-1/2), department scorecards, and caseload reallocations.
 Super Admin	<code>admin@setu.gov.in</code>	<code>/admin</code>	Platform-wide governance, live AI triage decision streams, master department taxonomy, forensic audit ledgers, and surge analytics.

* Default development password for all demo accounts is Password@123. One-click instant login is enabled on the Sign-In page.

4. 100% Native Multilingual System (12 Indian Languages)

The platform features zero-reload, instant reactive translation across 563 schema-synchronized keys with 100% parity across all 12 languages without mixed fallback text:

English
ENGLISH (EN)

தமிழ்
TAMIL (TA)

हिन्दी
HINDI (HI)

తెలుగు
TELUGU (TE)

ಕನ್ನಡ
KANNADA (KN)

മലയാളം
MALAYALAM (ML)

मराठी
MARATHI (MR)

বাংলা
BENGALI (BN)

ગુજરાતી
GUJARATI (GU)

ਪੰਜਾਬੀ
PUNJABI (PA)

ଓଡ଼ିଆ
ODIA (OR)

অসমীয়া
ASSAMESE (AS)

5. Key Platform Feature Modules

Geotagged Grievance Redressal

- Automated browser GPS auto-detection with ±meter accuracy.
- Instant camera & photo evidence upload (up to 10MB).
- 8-stage chronological redressal timeline with SLA countdown.

7-Step Service Application Wizard

- Direct apply for Piped Water, Smart Meters, Income Certificates, etc.
- Applicant identity, statutory enclosures, and pre-audit checklists.
- Instant official tracking reference generation (APP-2026-001).

Schemes & Eligibility Evaluator

- Repository of PM-KISAN, Ayushman Bharat, PMAY, SVANidhi, etc.
- 60-second rules-based eligibility checker matching age, income & land.
- Direct routing to official government portals.

SETU AI Assistant (Navigator)

- Verified floating AI government navigator in all 12 languages.
- Pre-populates application forms and generates document checklists.
- Live querying against official municipal service knowledge bases.

6. System Verification & Health Status

SUBSYSTEM	TEST VERIFICATION	STATUS
Multilingual Engine	Parity verification test across all 12 Indian language JSON bundles (563 keys).	100% PASS
Frontend Build	Vite production compilation transformed 1641 modules with 0 TypeScript errors.	COMPILED (0 ERRORS)
Backend API	REST API health endpoint GET /api/v1/health responding with 200 OK.	HEALTHY (200 OK)
PostgreSQL Database	Connected via Prisma ORM with master departments, services & demo user seeds.	CONNECTED